

Timolol Maleate Ophthalmic Solution USP 0.5% w/v

Description:
5 ml sterile clear colourless solution in a plastic bottle in a carton

Composition :
Each ml contains :
Timolol (as Timolol Maleate USP) 5mg
Benzalkonium Chloride NF 0.01% w/v
(as preservative)
In sterile aqueous buffered vehicle

Pharmacology :
 Iotim ED when applied topically in the eye has the action of reducing elevated as well as normal intraocular pressure, whether or not accompanied by glaucoma.
 The exact mechanism by which timolol reduces IOP has not been clearly defined. Fluorophotometric studies suggest that the reduced aqueous humor formation is the predominant effect.

Pharmacokinetics :
The onset of reduction in intraocular pressure following administration of Timolol can usually be detected within one half hours after a single dose; maximum effect usually occurs after one to two hours.

Pharmacodynamics :
Timolol is a competitive antagonist at both beta 1 and beta 2 adrenoceptors : it is therefore a non-selective beta blocking drug. Timolol combines reversibly with the beta-adrenergic receptor, and this inhibits the usual biologic response that would occur with stimulation of that receptor. This specific competitive antagonism blocks stimulation of the beta-adrenergic stimulating activity, whether these originate from an endogenous or exogenous source.

Indication :
lotim ED is indicated in the treatment of elevated intraocular pressure in patients with ocular hypertension or open angle glaucoma.

Recommended Dosage :
Instil one drop in the affected eye(s) twice daily or as directed by the physician.

Transfer from other agents :
When another topical beta blocking agent is being used, discontinue its use after a full day of therapy and start treatment with Iotim the next day with one drop of 0.25% Iotim ED in each affected eye twice a day. The dosage may be increased to one drop of 0.5% solution in each affected eye twice a day if the response is not adequate.

When transferring a patient from a single anti glaucoma agent other than a topical beta blocking agent, continue the agent and add one drop of 0.25% lotim ED in each affected eye twice a day. On the following day, discontinue the previous agent completely, and continue with lotim ED. If a higher dosage of lotim ED is required, substitute one drop of 0.5% solution in each affected eye twice a day.

Precaution / Warnings :

(a) **Precautions :**
 Lotim ED should be used during pregnancy and lactation only if the potential benefit justifies the potential risk to the fetus. Because of the potential for serious adverse reactions from lotim ED in nursing infants, a decision should be taken whether to discontinue nursing or to discontinue the drug taking into account the importance of the drug for the mother. Safety and effectiveness in pediatric patients have not been established.

(b) **Warnings:** As with many topically applied ophthalmic drugs, this drug is absorbed systemically. The same adverse reactions found with systemic administration of beta-adrenergic blocking agents may occur with topical administration. For e.g. severe respiratory reactions and cardiac reactions including death due to bronchospasm in patients with asthma, and rarely death in association with cardiac failure have been reported following systemic or ophthalmic administration of Timolol.

Cardiac Failure: Sympathetic stimulation may be essential for support of the circulation in individuals with diminished myocardial contractivity, and its inhibition by beta-adrenergic receptor blockade may precipitate more severe failure.

In patients without a history of cardiac failure continued depression of the myocardium with beta-blocking agents over a period of time can, in some cases, lead to cardiac failure.

Patients with chronic obstructive pulmonary disease (e.g. bronchitis, emphysema) of mild and moderate severity, bronchospastic disease, or a history of bronchospastic disease in which Timolol is contraindicated should, in general, not receive beta-blockers including lotim ED. Beta-adrenergic

Contraindications :

Timolol is contraindicated in patients with bronchial asthma, or a history of bronchial asthma or severe chronic obstructive pulmonary disease, sinus bradycardia; second or third degree atrioventricular block; overt cardiac failure; cardiogenic shock; hypersensitivity to any component of the product.

Side Effects / Adverse Reactions :
Local irritation is the only commonly observed adverse reaction. Bronchoconstriction is reported in elderly patients with pulmonary obstructive disease.

lotim ED is generally well tolerated following topical application to the eye

Signs and symptoms of ocular irritation including burning & stinging sensation, conjunctivitis, blepharitis, keratitis and blepharoposis have been reported occasionally in patients receiving topical timolol therapy. Visual disturbances including refractive changes (resulting from withdrawal of miotic therapy in some patients) have been infrequently associated with timolol therapy. Decreased corneal sensitivity, diplopia and ptosis have also occurred. Aggravation or precipitation of certain cardiovascular pulmonary and other disorders, presumably related to effects of systemic beta-adrenergic blockade, may occur during therapy with topical timolol and may include bradycardia, arrhythmia, congestive heart failure, hypotension, syncope, heart block, cerebrovascular accident, cerebral ischemia, palpitation, dyspnea, nasal congestion, cough bronchospasm (mainly in patients with pre-existing bronchospastic disease).

and respiratory failure. Headache, depression, dizziness, paresthesia, asthenia, diarrhoea and nausea have also been reported in patients receiving topical timolol therapy.

Drug interactions:

Although Timolol used alone little or no effect on pupil size, mydriasis resulting from concomitant therapy with Timolol and epinephrine has been reported occasionally. Close observation of the patient is recommended when a beta blocker is administered to patients receiving catecholamine-depleting drugs such as reserpine, because of possible additive effects and the production of hypotension and/or marked bradycardia, which may produce vertigo, syncope or postural hypotension. The concomitant use of beta-adrenergic blocking agents with digitalis and calcium antagonists may have additive effects in prolonging atrioventricular conduction time. Patients who are receiving beta blockers and oral or intravenous serotonin (5-HT₂) antagonists should be observed for potential additive effects on blood pressure, both systemic and on intraocular pressure. The concomitant use of two topical beta adrenergic blocking agents is not recommended.

Pregnancy & Lactation :

lotim ED should be used during pregnancy and lactation only if the potential benefit justifies the potential risk to the foetus. Because of the potential for serious adverse reactions from lotim ED in nursing infants, a decision should be taken either to discontinue nursing or discontinue the drug taking into account the importance of the drug for the mother. Safety and effectiveness in paediatric patients have not been established.

Symptoms and Treatment for Overdosage :

There have been reports of inadvertent overdosage of lotim ED resulting in systemic Beta-adrenergic blocking effects such as dizziness, headache, shortness of breath, bradycardia, bronchospasm and cardiac arrest.

Over dosage has been reported with Timolol Maleate Tablets. A 30 years old female ingested 650 mg of Timolol (maximum recommended oral daily dose is 60 mg) had experienced second and third degree heart block. She recovered without treatment but approximately two months later developed irregular heartbeat, hypertension, dizziness, linnitus, increased pulse rate and borderline first degree heart block. The oral LD 50 of the drug is 1190 and 900 mg / kg in female mice and female rats respectively.

An *in vitro* hemodialysis study, using ^{14}C timolol added to human plasma or whole blood, showed that timolol was readily dialyzed from these fluids; however, a study of patients with renal failure showed that timolol did not dialyze readily.

The most common signs and symptoms to be expected with overdosage with administration of a systemic beta-adrenergic receptor blocking agent are symptomatic bradycardia, hypotension, bronchospasm and acute cardiac failure. The following therapeutic measures should be considered.

1. **Gastric Lavage:** If ingested
2. **Symptomatic bradycardia:** Use atropine sulfate intravenously in a dosage of 0.25mg to 2mg to induce vagal blockade. If bradycardia persists, intravenous isoproterenol hydrochloride should be administered cautiously. In refractory cases, the use of a transvenous cardiac pacemaker may be considered.
3. **Hypotension:** Use sympathomimetic pressor drug therapy such as dopamine, dobutamine or levaterenol. In refractory cases, the use of glucagon hydrochloride has been reported to be useful.
4. **Use isoproterenol hydrochloride:** Additional therapy with aminophylline may be considered.
5. **Acute cardiac failure:** Conventional therapy with digitalis, diuretics and oxygen should be instituted immediately. In refractory cases, the use of intravenous aminophylline is suggested. This may be followed if necessary by glucagons hydrochloride which has been reported to be useful.
6. **Heart block (second or third degree):** Use isoproterenol hydrochloride or a transvenous cardiac pacemaker.

Effects on Ability to Drive and Use Machine :

This medicine may cause blurred vision after application even at normal dosages and with proper use. This can subsequently impair reaction time while driving or operating machinery.

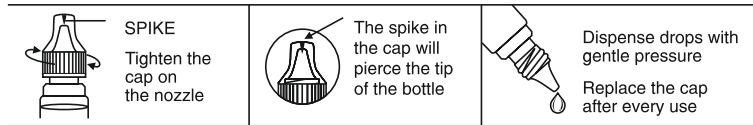
Pack Size :
5ml in plastic bottle in a carton.

Storage :

Keep medicine away from the reach of children.
For opened and Unopened bottle : Store below 30°C. Protect from light.
Use the solution within 28 days after opening the bottle.
Expiry date 24 months from the manufactured date.
Route of administration: For Ophthalmic Use

MAL.: 19990571AZ

MODE OF USE



Manufactured in India by :

FDC Limited
B-8, MIDC Industrial Area, Waluj,
Chhatrapati Sambhajnagar - 431 136,
Maharashtra.

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Imported by
UNIMED SDN BHD
53, Jalan Tembaga SD 5/2B
Bandar Sri Damansara,
52200 Kuala Lumpur, Malaysia.

KEMENTERIAN KESIHATAN MALAYSIA

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FDC Limited	LEAFLET	Minipharma Code No.: Front: 2306 / Back: 2307	
	Size	110 (L) x 144 (H) mm	
Folded Size	-----		
No. of Folds	-----		
GSM	60 gsm		
Paper	Maplitho Paper		
Specification No.			
Colour	■ BLACK		
SAP Code: D1	Supercedes SAP Code: 2000006907		
Location: Waluj (Malaysia)	Prepared on: 15-12-2025		
Reason for Change:			
1) & word removed from storage.			
2) In address 'Aurangabad' replaced with "Chhatrapati Sambhajnagar".			